forward SDE

$$\begin{cases} dx_t = f(x_t, t)\,dt + g(t)\,dW_t \\ \dfrac{\partial p_t(x)}{\partial t} = -\nabla \cdot [f(x,t)\,p_t(x)] + \frac{1}{2} g(t)^2 \Delta p_t(x) \end{cases}$$

---

let reverse time $\tau = T - t$

reverse SDE

$$\begin{cases} dx_\tau = \tilde{f}(x_\tau, t)\,\underbrace{d\tau}_{-dt} + g(t)\,\underbrace{d\bar{W}_\tau}_{\approx dW_t} \\ \dfrac{\partial p_t(x)}{\underbrace{\partial \tau}_{-\partial t}} = -\nabla \cdot [f(x,t)\,p_t(x)] + \frac{1}{2} g(t)^2 \Delta p_t(x) \end{cases}$$

Matching distribution

$$\underbrace{-\nabla \cdot (fp) + \tfrac{1}{2} g^2 \Delta p}_{\text{forward}} = \underbrace{\nabla \cdot (\tilde{f} p) - \tfrac{1}{2} g^2 \Delta p}_{\text{reverse}}$$

$$\nabla \cdot \left((f + \tilde{f})\, p_t(x) \right) = g(t)^2 \Delta p_t(x)$$

$$= \nabla \cdot \left[g(t)^2 p_t(x) \nabla \log p_t(x) \right]$$

then $\tilde{f}(x,t) = -f(x,t) + g(t)^2 \nabla \log p_t(x)$

---

hence,

$$dx_t = \Big[f(x_t, t) - \underbrace{g(t)^2 \nabla \log p_t(x)}_{\text{고밀도 지역}} \Big] dt + g(t)\,dW_t$$

$$x_{t-\Delta t} = x_t + \Big[-f(x_t, t) + g(t)^2 \nabla \log p_t(x) \Big] \Delta t + g(t) \sqrt{\Delta t}\, Z$$

OU Process

$$\frac{\partial p}{\partial t} = -\nabla \cdot (fp) + \frac{1}{2}\sigma^2 \nabla^2 p^2$$

① $dX_t = -\beta X_t dt + \sigma dW_t$

$$\begin{cases} p = x^3 + y^2 \\ \nabla p = (3x^2, 2y) \\ \nabla \cdot p = 3x^2 + 2y \\ \nabla^2 p = 6x + 2 \end{cases}$$

② $X_t | X_0 \sim N(e^{-\beta t} X_0, \sigma_t^2 I)$

(i) $\sigma_t^2 = \frac{\sigma^2}{2\beta}(1 - e^{-2\beta t})$

(ii) $X_t | X_0 \sim N(0, \frac{\sigma^2}{2\beta} I)$ when $t \to 0$

③ $d\overline{X_t} = (-\beta X_t - \sigma^2 \nabla_x \log P_t) dt + \sigma d\overline{W_t}$

$d\overline{X_t} = (f(\overline{X_t}, t) - g^2(t) \nabla_x \log P_t(\overline{X_t})) dt + g(t) d\overline{W_t}$ (ODE) "흐름, 다양성"

$d\overline{X_t} = (f(\overline{X_t}, t) - \frac{g^2(t)}{2} \nabla_x \log P_t(\overline{X_t})) dt$ (SDE) "노이즈, 분산"

$\Rightarrow$ same P_t (Fokker - Planck)

$\nabla_x \log_t P \approx S_\theta(x, t)$

$$\mathcal{L}(\theta) = \int_0^T \lambda(t) \mathop{\mathbb{E}}_{X_t \sim P_t} \left[\| S_\theta(X_t, t) - \nabla_{x_t} \log p_t(X_t) \|^2 \right] dt$$

$$\begin{cases} \mathcal{L}_{DSM}(\theta) = \int_0^T \lambda(t) \, \mathbb{E}_{X_0} \mathbb{E}_{X_t|X_0} \left[\left\| S_\theta(X_t, t) - \nabla_{X_t} \log P_{t|0}(X_t|X_0) \right\|^2 \right] dt \\ \mathcal{L}_{SSM}(\theta) = \int_0^T \lambda(t) \mathop{\mathbb{E}}_{X_t} \left[\| S_\theta(X_t, t) \|^2 + 2 \mathop{\mathbb{E}}_{v} \left[\frac{d}{du} v^T S_\theta(X_t + hv, t) \Big|_{h=0} \right] \right] dt. \end{cases}$$

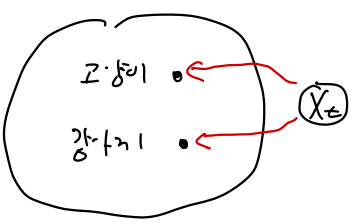
---

① $\nabla_{x_t} \log P_t(x_t) = \dfrac{\nabla_x P_t(x)}{P_t(x)} = \displaystyle\int \nabla_{x_t} P_{t|0}(X_t|X_0) \frac{P_0(X_0)}{P_t(X_t)} \, dX_0$

그럴듯한 그림

$$= \int \nabla_{x_t} \log P_{t|0}(X_t|X_0) \cdot \underbrace{\frac{P_0(X_0)}{P_t(X_t)} \cdot P_{t|0}(X_t|X_0)}_{= P_{0|t}(X_0|X_t)} \, dX_0$$

그럴듯한 그림



$$= \mathop{\mathbb{E}}_{X_0|X_t} \left[\nabla_{X_t} \log P_{t|0}(X_t|X_0) \right]$$

고양이 그림.
강아지 그림 .... 평균

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② $\mathcal{L}_{DSM}(\theta) = \displaystyle\int_0^T \lambda(t) \mathop{\mathbb{E}}_{X_t} \left[\| S_\theta(X_t, t) - \nabla_{x_t} \log p_t(X_t) \|^2 \right] dt$

$$= \lambda(t) \mathop{\mathbb{E}}_{X_t} \left[\|S_\theta\|^2 - 2 \langle S_\theta, \nabla_{X_t} \log P_t(X_t) \rangle \right] + \| \nabla \cancel{\log} P_t \|^2 \Big]$$

$$= \lambda(t) \mathop{\mathbb{E}}_{X_t, X_0} \left[\|S_\theta\|^2 - 2 \langle S_\theta, \nabla_{X_t} \log P(X_t|X_0) \rangle \right]$$

$$= \lambda(t) \mathop{\mathbb{E}}_{X_0} \mathop{\mathbb{E}}_{X_t|X_0} \left[\| S_\theta - \nabla_{X_t} \log P(X_t|X_0) \|^2 + \cancel{C} \right]$$

① $\mathcal{L}(\theta) = \int_0^T \lambda(t) \mathop{\mathbb{E}}_{x_t} \left[\| s_\theta(x_t, t) - \nabla_{x_t} \log p_t(x_t) \|^2 \right] dt$

$$\Downarrow$$

$$\|s_\theta\|^2 - 2 \langle s_\theta, \nabla \log p_t \rangle + \cancel{\| \nabla \log p_t \|^2}$$

$\langle s_\theta, \nabla \log p_t \rangle$ ↑

를 위해서 하는데,

모든 각도 v에서의

$\langle \cdot, \cdot \rangle$ 없다.

---

② $\mathop{\mathbb{E}}_{x_t} \left[\langle s_\theta, \nabla \log p_t \rangle \right]$

Note : 부분적분. (치환)

$= \int s_\theta(x)^T \cdot \nabla p_t(x)\, dx \qquad \left(\nabla \log p_t = \frac{\nabla p_t}{p_t} \right)$

$\cancel{\left[s_\theta(x) \cdot p(x) \right]_{-\infty}^{\infty}}$

$- \int_{-\infty}^{\infty} s_\theta'(x) \cdot p(x)\, dx.$

$= - \int (\nabla \cdot s_\theta(x))\, p_t(x)\, dx$ (부분적분)

$= - \mathop{\mathbb{E}}_{x_t} \left[\nabla \cdot s_\theta(x, t) \right]$ (Jacobian)

$= \mathrm{Tr}(D_x s_\theta(x, t))$

---

③ $\mathrm{Tr}(D_x s_\theta) = \mathop{\mathbb{E}}_{v} \left[v^T (D_x s_\theta) v \right]$

when $v \sim N(0, I)$

$\left(\mathop{\mathbb{E}}_{v} \left[\sum_{i,j} A_{ij} v_i v_j \right] \right)$

trace만 생존

---

④ $v^T (D_x s_\theta v) = v^T \left(\frac{d}{dh} s_\theta(x + hv, t) \Big|_{h=0} \right)$

---

⑤ $\mathcal{L}_{SSM}(\theta) = \int_0^T \lambda(t) \mathop{\mathbb{E}}_{x_t, v} \left[\| s_\theta(x_t, t) \|^2 + 2 \frac{d}{dh} v^T s_\theta(x_t + hv, t) \Big|_{h=0} \right] dt.$

- OU Process : $dX_t = -\beta X_t dt + \sigma dW_t$

- $X_t | X_0 \sim N(\mu_t, \Sigma_t)$

- $X_t = \gamma_t X_0 + \sigma_t \varepsilon \sim N(0, I)$

$$\boxed{\begin{aligned} \mu_t &= \gamma_t X_0 \\ &= e^{-\beta t} X_0 \end{aligned} \qquad \begin{aligned} \Sigma_t &= \sigma_t^2 I \\ &= \frac{\sigma^2}{2\beta}(1 - e^{-2\beta t}) I \end{aligned}}$$

Score func

$\nabla_{x_t} \log p(X_t | X_0)$

$$= -\frac{X_t - \gamma_t X_0}{\sigma_t^2} = -\frac{\varepsilon}{\sigma_t}$$

$\because$ $$p(x | \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \cdot \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

- let $s_\theta = -\dfrac{\varepsilon_\theta(x, t)}{\sigma_t}$

then $$\mathcal{L}(\theta) = \int_0^T \cancel{\frac{\lambda(t)}{\sigma_t^2}} \mathop{\mathbb{E}}_{x_0, \varepsilon} \left[\left\| \varepsilon_\theta(\gamma_t X_0 + \sigma_t \varepsilon, t) - \varepsilon \right\|^2 \right] dt \quad . \qquad (DSM)$$

---

① sample $t \sim \text{Uniform}(0, T)$

② " $x_0 \sim p_{data}$.

③ " $\varepsilon_0 \sim N(0, I)$

④ compute $X_t = \gamma_t X_0 + \sigma_t \varepsilon$

⑤ " $\mathcal{L}_\theta = \| \varepsilon_\theta(X_t, t) - \varepsilon \|^2$

⑥ update θ

Note. Reverse SDE

$$d\overline{X_t} = \left[\frac{\sigma^2}{\sigma_t} \cdot \varepsilon_\theta(\overline{X_t}, t) - \beta \overline{X_t} \right] dt + d\overline{W_t}$$